Paul Levitt

Technical Architect

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Creative Problem Solver | Strong Communicator | Avid Learner and Mentor

Technical Architect with deep expertise in Node.js, Docker, and internationalization; specializing in designing and inventing metadata languages for enterprise scale no-code platforms. Backed by strong foundations in distributed systems, cloud development (AWS), CI/CD pipelines, No/SQL databases, and interpersonal communication, I create scalable, reliable metadata driven PaaS architectures used to deliver meaningful, innovative customer-first SaaS products that matter. I take pride in having a strong sense of ownership, a focus on innovative and simple solutions, clean maintainable source, documentation, and great developer experience. Passionate about continued learning and seeking growth, being a helpful approachable resource and mentor, finding enjoyment in everything, and Lego *(lots of Lego!)*

WORK EXPERIENCE

Lifion by ADP - *New York, NY* January 2018 to July 2024

Technical Architect

Leading multiple teams, features and cross-functional initiatives, driving the architectural direction and roadmap of ADP's no-code platform,Lifion; whilst delivering functionality and features enabling global enterprise clients and multi-cultural users of Lyric (ADP's flagship next-generation HCM SaaS, Lyric)

- Integrated MyADP's Things To Do micro-frontend and enhanced internal API's, centralizing task management from multiple ADP products, whilst enabling rich user interactions for Lifion's tasks *(Node.js, React v16, Angular v17, GoLang)*
- Engineered runtime caching mechanisms for user authentication of search data, delivering substantial performance improvements of ~65% in search request-response cycles whilst eliminating search data reindexing *(Node.js, Python, SOLR)*
- Led workshops and provided training materials educating cross-functional product & platform teams on localization capabilities and usage; ensuring Lifion can correctly process data for all of ADP's 140 operating countries
- Led workshops, provided training materials, and assisted platform teams implementing features for localisable metadata, translatable metadata & code resources and formatting of culturally aware data
- Led a team building a Translations Management System, enabling product to self-manage supported languages and automate the physical translation of string resources (stored in either metadata and code)
- Developed a self-service system for sales teams to manage demo client databases, transforming the prospective client presentation process from single-instance limitations to multiple concurrent
- Contributed to the development of ADP Assist's initial implementation in ADP Mobile, providing a unified request/response schema allowing a single chatbot-like end user interaction across multiple disparate ADP solutions.
- Created metadata driven Chatbot solution for product teams to build conversations appropriate to their domain; for example, "Book time off" for the Time & Attendance team(s). Accessible via multiple channels (Text messages, voice assistants, Messenger, etc.) including integration with MyADP's Assist chat infrastructure providing access to functionality from other ADP products *(Node.js, AWS Lambda, S3 and DynamoDB)*

Lifion by ADP - *Bristol, North Somerset, UK* February 2010 to December 2017

Technical Architect

Lead technical architect and co-founder of Lifion by ADP, scaling and leading multiple development teams based in the UK, US & India.

- Spearheaded internationalization initiative; delivered multiple modules enabling formatting of culturally-aware data covering 283 regions *(Node.js)*

- Engineered obfuscation mechanism for metadata collections, enabling app bundling, cache reuse and reduced Time To Interact *(Node.js)*

- Invented unified metadata superset - "DNA" - replacing disparate metadata type structures under single DSL, and replacing multiple type-specific services, streamlining platform operations through a unified compiler *(Node.js)*

- Created data-structures and algorithms for layering metadata, providing flexible and deterministic localizations, cross-feature integrations, client customisations, and more

- Created metadata types and services enabling the drag & drop design of business rules, enabling validations, computed data and UI interactions, removing the need for custom frontend JavaScript and most backend plugins, which in-turn facilitated near cost-less upgrades of clients with customizations (offered free by SLA) *(C#, JavaScript, SQL Server)*

- Contributed to conceptual, logical and physical data models for ADP's "Global Person" initiative, enabling better cross-region data integrations, cross-employment history, reducing background-check costs, durations and increasing employment statistics data accuracy (used by US & other government agencies)

- As part of ADP's Enterprise Architecture committee, contributed to multiple conceptual designs for future company wide initiatives and investigations

- Delivered functionality providing pluggable data sources providing metadata driven integrations with internal/external systems, for example; Providing clients access to internal metadata capabilities for self-service customisation, and embedding data from external REST API's systems *(VB.net, C#)*

Vizual Business Tools PLC - *Worle, Somerset, UK* November 2007 to February 2010

Software Engineer

Software engineer on the Core Architecture Team enhancing our low-code platform, leading multiple teams in India, and designing and building HCM product features built upon our platform

- Design, development and support of a no-code platform used to build Human Resource Management products *(VB.net, C#, SQL Server, JavaScript)*

- Architected and implemented extensible workflow design and processing capabilities in out low-code platform, enabling customizable business processes and enhanced system flexibility *(C#)*

- Developed automated system to generate documentation from product metadata, streamlining knowledge management and reducing documentation overhead *(VB.net)*

Benchmark Software Ltd. - *Street, Somerset, UK* July 1998 to November 2007

Research and Development Director

Software engineer and company director responsible for the ongoing development of Benchmark's accounting & bookkeeping, Payroll, CRM and bespoke solutions for clients in the UK and Ireland.

- Ownership and technical oversight of multiple core ledgers and add-ons including the Nominal ledger, Credit control, Sales Order Processing, Purchase Order Processing, Sales Quotes and Job Invoicing. *(Visual Basic 6, C++, Embedded C)*

- 0 to 1 development and management of UK Payroll solution, full legislative HMRC compliant online filing of employer & employee TAX and National Insurance contributions. Fully integrated with accounting & booking keeping products. *(VB.net & C#, Crystal Reports, REST API's)*

- 0 to 1 development and management of Customer Relationship Management solution; Initially an internal tool for our Support and Sales team members, later released as new product offering
  (Visual Basic 6, Crystal Reports, telephony, REST API's)

- Frequent communication with clients, both prospective and existing. From onboarding, installation and training of existing solutions; to requirements gathering, solutioning, design, quotations/bill of quantities, and implementation of bespoke solutions.

EDUCATION

University of the West of England - *Bristol, North Somerset, UK* *July 2000*

Bachelors of Science in Software Engineering

Strode College - *Street, Somerset, UK* *July 1995*

BTEC National Diploma in Electrical and Electronic Engineering

PROFESSIONAL SKILLS

- Node.js, Typescript, C#, VB.net, Python, VB6, Go
- React, Angular
- AWS, Docker, Kubernetes, Github, Stash, Jenkins
- Confluence, Jira, Rally, Slack, Teams

VOLUNTEER EXPERIENCE

- **Volunteer, K-280 Pre-K Parent Advisory Committee** - *Brooklyn, NY* 2024 - 2025
- **Parent Liaison, PS-10 Extended Day Executive Board** - *Brooklyn, NY* 2025 - *Present*

AWARDS AND HONORS

“Dynamically composing products using capsules” US Patent # US-10558436-B2 - *New York, NY* February 2020

Abstract: A method and apparatus for dynamically composing applications is presented. A computer system identifies a set of parameters based on user input. The computer system then activates a portion of a plurality of capsules stored in a data storage to form a set of activated capsules based on the set of parameters. Further, the computer composes an application in real-time using a base capsule and the set of activated capsules in which the application is specifically tailored for a particular state. The application and the plurality of capsules take up a reduced amount of memory space.